

SymPy Bug Card

Bug 46: `Matrix.is_diagonalizable` Ignores Eigenvalue Collision

Task. Decide whether the symbolic matrix is diagonalizable.

$$M = \begin{pmatrix} a & 1 \\ 0 & b \end{pmatrix}.$$

SymPy's answer.

$$\text{is_diagonalizable}(M) = \text{True}, \quad P = \begin{pmatrix} 1 & -1/(a-b) \\ 0 & 1 \end{pmatrix}.$$

Correct answer. The answer is conditional: M is diagonalizable for $a \neq b$, but it is not diagonalizable when $a = b$.

Explanation. Substituting $b = a$ gives $\begin{pmatrix} a & 1 \\ 0 & a \end{pmatrix}$, a nontrivial Jordan block. Its only eigenspace is one-dimensional while the eigenvalue a has algebraic multiplicity 2, so an unconditional True is wrong.

Likely root cause. In `sympy/matrices/eigen.py`, SymPy trusts generic `eigenvects()` output and checks the symbolic eigenvalue keys a and b independently, without carrying the side condition $a \neq b$ introduced by $1/(a-b)$. This is a new instance of a known failure mode.

Artifact (GitHub).

[bug-report/bug-046-matrix-is-diagonalizable-ignores-eigenvalue-collision/](#)